

AI Methodology Appendix

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Part 7: AI Methodology Appendix (P.R.I.M.E.)

Entry 1: Dataset Selection and Causal Question Framing

- **Prompt:** "I need to pick a causal question and dataset for my ECON 5200 final project using Double Machine Learning. Needs real data, 1,000+ observations, clear treatment and outcome, and a compelling selection bias story. Recommend one."
- **Response:** Claude recommended the 401(k) eligibility dataset from Abadie (2003) and Chernozhukov et al. (2018), available via `doubleml.datasets.fetch_401K`. Rationale: canonical DML demonstration dataset, N=9,915, binary treatment, policy-relevant outcome, large income confound makes the case for DML pedagogically clear and grader-compelling.
- **Iterate:** Asked Claude to verify the dataset loaded correctly in the current environment and confirm variable names before committing to the topic.
- **Modify:** Added `hown` (homeownership) as an additional control beyond the standard specification — captures wealth-driven savings capacity partially independent of income.
- **Evaluate:** Ran `fetch_401K(return_type='DataFrame')` in the kernel, confirmed N=9,915, confirmed variable names, verified balance check showed expected income gap (~\$15,368). Dataset confirmed viable.

Entry 2: DML Implementation and Debugging

- **Prompt:** "Write Python code to estimate the causal ATE of 401(k) eligibility on `net_tfa` using DoubleMLPLR with GradientBoostingRegressor nuisance models and 5-fold cross-fitting. Extract SE and 95% CI. Then add a robustness check with `RandomForestRegressor`."
- **Response:** Claude provided working code using `doubleml.DoubleMLPLR` with `score='partialling out'`, extracting `plr.coef`, `plr.se`, and `plr.confint()` for inference. It flagged that `LinearDML` from `econml` returns an empty `coef_` array when `X=None`, recommending `DoubleML` instead for clean ATE inference.
- **Iterate:** Initial code used `n_estimators=100`; encountered `UserWarning` from `econml` about discrete treatment — resolved by switching to `doubleml` package. Also fixed

`fetch_401K(return_X_y=False) → fetch_401K(return_type='DataFrame')` after API mismatch.

- **Modify:** Added `max_depth=3` and `learning_rate=0.05` to GBM for regularization; added `max_features='sqrt'` to RF.
 - **Evaluate:** Ran both models in kernel. GBM ATE = \$9,581 (SE=\$1,479), RF ATE = \$8,984 (SE=\$1,315). Difference ~\$597, within one SE — robust. Naive OLS (\$19,559) confirmed substantially higher, consistent with expected upward bias.
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Entry 3: Threats to Identification and Technical Report

- **Prompt:** "Write the Threats to Identification section for this DML analysis. 500+ words, three specific threats, direction of bias for each, what research design would address each. Be honest — graded on critical thinking, not on selling the result."
- **Response:** Claude identified three threats: (1) worker-employer sorting on unobserved savings preferences (most serious; upward bias); (2) savings crowd-out across vehicles and SIPP measurement limitations (ambiguous); (3) firm-level compensation bundling beyond income (upward). For each it specified the identification strategy that would resolve it.
- **Iterate:** Asked Claude to add quantitative framing to the external validity concern — specifically how much residual confounding would be needed to make the estimate zero (answer: ~75% of the DML estimate).
- **Modify:** Added explicit mention of SECURE 2.0 Act and automatic enrollment as reasons external validity is limited beyond the 1990s SIPP sample.
- **Evaluate:** Reviewed each threat against the actual dataset. Balance check confirms the income gap, supporting threat 1. `pira` is in the dataset and controlled for, partially mitigating threat 2. Threat 3 is unverifiable from SIPP alone — correctly flagged as what we cannot rule out. Human judgment confirmed: honest, specific, does not oversell the result.